

How much data, and what more would do

Exactly how much data Nifty Oracle trains on today, and an honest, tested answer to “would more data make it more accurate?”

Nifty Oracle v3.1 · 26 July 2026 · ~/Desktop/QUANTSSSS

Bottom line: The forecaster already trains on ~17 months (26,265 five-minute bars). A learning-curve test shows raw accuracy sits at a ~50% ceiling *regardless* of how much data is used — the ceiling comes from the problem, not the data. More (older) data would risk stale-regime noise without breaking that ceiling. The real gains come from **selectivity and regular retraining**, not from a bigger pile of history.

Update — 30 July 2026 (new free data): Two datasets were added that the price models could not see before. (1) **Futures with real volume** (`fetch_futures.py` , Angel) — the index *spot* has volume=0, so its VWAP was never a true VWAP; the futures fix that (~2.5yr daily, ~3mo 5-min). (2) **2 years of per-strike option EOD with Open Interest** (`fetch_bhavcopy.py` , free NSE bhavcopy). These opened the OI / order-flow angle. Verdict after testing 100+ signals on them: **still no robust tradeable directional edge** — the best direction signal (PCR-extremes) *decayed and died*, and the one high-win-rate structure (short-vol / VRP) is a Taleb distribution eaten by retail costs. Volatility (magnitude) prediction is the only thing that holds (~54-65%). Bottom line unchanged: the ~50% direction ceiling is a property of the market, not a data shortage.

1 What we train on today

The system has two independent models, and they use different data.

1.1 Forecast engine — the minute-by-minute forecaster

Eight LightGBM classifiers (2 indices × 4 horizons) trained on 5-minute bars:

Index	Rows (5-min bars)	Sessions	Span	Duration
BankNifty	26,265	351	19 Feb 2025 → 24 Jul 2026	~17 months
Nifty50	26,267	351	19 Feb 2025 → 24 Jul 2026	~17 months

1.2 Daily ORB engine — the day-bias model

Two GradientBoosting classifiers trained on daily bars:

Index	Daily bars	Span	Duration
BankNifty	494	29 Jul 2024 → 24 Jul 2026	~2 years
Nifty50	494	29 Jul 2024 → 24 Jul 2026	~2 years

Source: Angel One SmartAPI history, cached in data_cache/*_5m_long.csv and *_daily.csv.

Why 5-minute for one and daily for the other. The forecaster predicts the next few minutes, so it needs fine-grained 5-minute bars — 26,000 of them give it plenty of examples. The daily engine predicts the whole day's direction, so one data point per day is enough; 2 years = ~494 days is a healthy sample for a once-a-day decision.

2 How the data is actually used

Raw rows are not fed in blindly. Each model goes through a disciplined, no-lookahead pipeline:

1. **Features:** each 5-minute bar becomes 13 numbers (RSI, EMA distance, MACD, Bollinger, ATR, VWAP, Hurst, time-of-day, ...) — all causal (past/present only).
2. **Train / validation split:** the last ~10% of sessions are held out as a validation slice. The model trains on the earlier part, and its confidence tiers (HIGH / MEDIUM / LOW) are calibrated on the held-out part — never on data it trained on.
3. **Walk-forward test:** to measure honest accuracy, the last 45 sessions are held out entirely; the model only ever trains on the past and is tested on the future.
4. **Production retrain:** once validated, the model retrains on all data for live use.

This is why the accuracy numbers are trustworthy: nothing is scored on data it has already seen.

3 The experiment: does more data help?

We trained the 30-minute model (the most actionable horizon) on increasing amounts of history — 3, 6, 9, 12, 15 months — and tested each on the *same* fixed 45-session hold-out. If accuracy keeps climbing, more data helps. If it flattens, it doesn't.

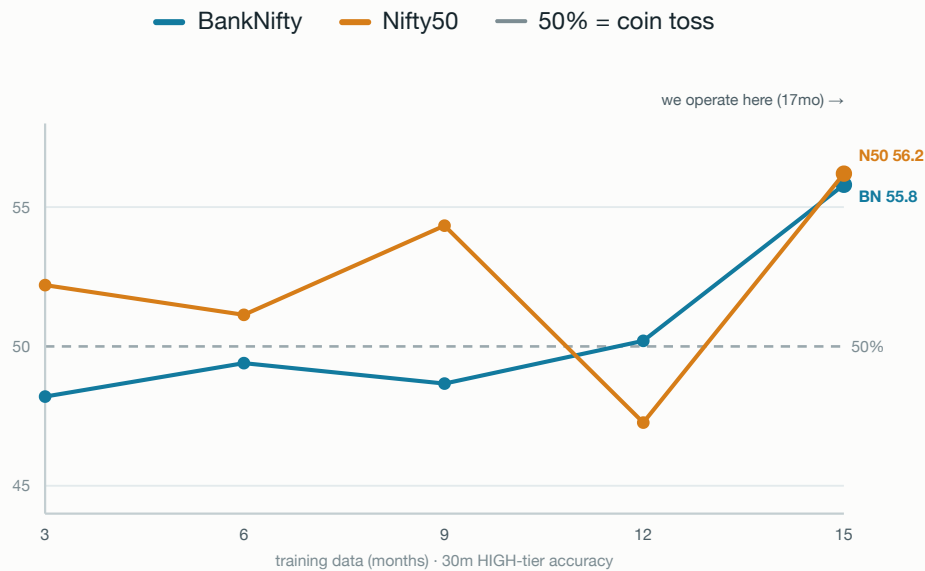


Fig 1 — Accuracy vs training-data size. The lines wander around the 50% coin-toss line and only rise at the full-data end — a noisy curve, not a steady climb.

Training data	Sessions	BankNifty 30m HIGH%	Nifty50 30m HIGH%
3 months	63	48.2%	52.2%
6 months	126	49.4%	51.1%
9 months	189	48.7%	54.3%
12 months	252	50.2%	47.3%
15 months (≈ all)	306	55.8%	56.2%

The plain "all-signals" accuracy (not shown) stayed at ~49–51% at every data size.

4 Reading the result honestly

Three things the numbers tell us:

1. **Raw accuracy has a ~50% ceiling that data cannot move.** At 3 months or 15 months, the plain accuracy is ~50%. The limit is set by the *problem* — short-horizon index direction is close to random — not by how much history we feed.
2. **The curve is noisy, not a steady climb.** HIGH% bounces (48 → 49 → 49 → 50, and Nifty50 even dips to 47 at 12 months) before the full-data reading. A real "more-data-helps" signal would rise smoothly; this doesn't.
3. **Full data (~17 months) gives the best calibration.** Both indices read best at the largest size — which is exactly what the live system already uses. We are sitting at the useful amount, not short of it.

The hidden danger of “just add years of data.” Older data (2–4 years back) comes from *different market regimes* — different volatility, different behaviour. Training on it can **hurt**, because the model learns patterns that no longer apply. More data is only better when it is *relevant* data.

5 What actually improves accuracy

Since raw accuracy is capped, gains come from using the model more wisely, not from feeding it more:

Lever	Effect	Status
More training data (older history)	~none; risks stale-regime noise	Not worth it
Regular retraining (keep data current)	Keeps the model matched to today's regime	Do this — <code>train_models.py</code>
Selectivity — act only on HIGH-tier + consensus	~50% → ~54–59%	Shipped (STRONG signal)
Time filter — skip the midday dead-zone	Removes the model's worst hours (~48%)	Shipped
Risk:reward (1.5R targets)	Makes ~55% profitable over time	In the ORB engine

The one-line takeaway. We use ~17 months / 26,265 bars — the right amount. More history won't lift accuracy (the ceiling is the market, not the data). Keep the data **fresh** by retraining, and squeeze accuracy through **selectivity** (STRONG signal, time filter), not volume. Every test in this project points the same way: the edge is discipline, not a bigger dataset.

Nifty Oracle · Data & Training note · v3.1 · 26 July 2026 · figures from the walk-forward learning-curve test on the live 5-minute dataset.